

Investigating Determinants of Birth Weight Using Bayesian Tree-Based Non-Parametric Modeling

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Why Low Birth Weight Matters

- ▶ LBW (≤ 2.5 kg) is linked to higher neonatal mortality and life-long morbidity.
- ▶ Incidence is driven by a complex mix of genetic, socioeconomic, behavioral, and environmental factors.
- ▶ Public-health goal: pinpoint *interpretable* high-risk subgroups for targeted intervention.

Gaps in Existing Approaches

- ▶ **Classical regression**: additive; under-represents interactions.
- ▶ **CART** trees: interpretable but ignore full birth-weight distribution.
- ▶ **Bayesian density models**: flexible yet computationally heavy and opaque.
- ▶ Proposal: combine CART interpretability with Bayesian uncertainty by using a *Dirichlet–Multinomial* (DM) marginal likelihood as the splitting criterion.

2021 U.S. Natality Data

- ▶ 3.6 M birth-weight records across all 50 states.
- ▶ Seven binary predictors: race, smoking, marital status, maternal age (above/below 33), High School completion, prenatal care adequacy, infant sex.
- ▶ All $2^7 = 128$ predictor combinations $\Rightarrow$ 128 classes.

Table: Binary predictor definitions

Label	Natality field	Value = 1	Value = 0
Boy	sex	Infant is male ("M")	Infant is female ("F")
Married	dmar	Mother is married	Mother not married
Black	mrace15	Black / African American	Any other race
Over33	mager	Maternal age > 33 yr	Maternal age $\leq$ 33 yr
HighSchool	medu	High-school education completed	Otherwise
FullPrenatal	prenatal	Adequate prenatal care	Inadequate / none
Smoker	cig_0	Any prenatal smoking	No smoking

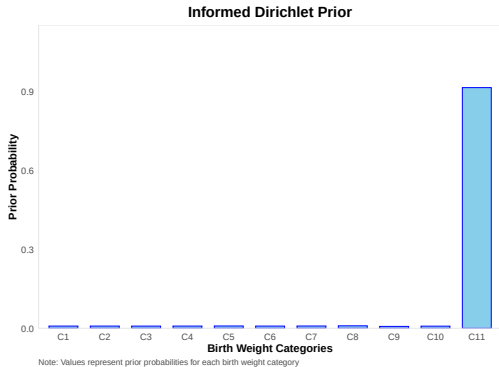
Birth-Weight Binning

- ▶ Ten LBW increasing birth-weight categories (0–2.5 kg) plus one normal-weight category.
- ▶ Dirichlet hyperparameters set using 2020 frequencies to avoid double-dipping.
- ▶ Final data: 128×11 ("full") and 128×10 ("LBW-only") count matrices.

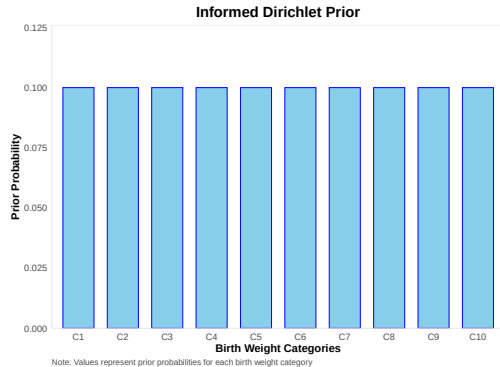
Table: Birth-weight quantile cut-points and Dirichlet priors

Quantile	LBW + Normal		LBW-only	
	Range (g)	Prior (%)	Range (g)	Prior (%)
Q1	227–1170	0.84	227–1170	10
Q2	1170–1644	0.84	1170–1644	10
Q3	1644–1899	0.83	1644–1899	10
Q4	1899–2069	0.83	1899–2069	10
Q5	2069–2183	0.87	2069–2183	10
Q6	2183–2270	0.83	2183–2270	10
Q7	2270–2350	0.86	2270–2350	10
Q8	2350–2410	0.93	2350–2410	10
Q9	2410–2460	0.71	2410–2460	10
Q10	2460–2500	0.80	2460–2500	10
Normal	>2500	91.67	—	—

Dirichlet–Prior



Full model prior: LBW + NBW



LBW-only prior: LBW

Comparison of informed Dirichlet hyperparameters (2020 quantiles)

From Multinomial Counts to a DM Node Model

$\mathbf{x}_i$ = dummy-encoded predictor vec. for class i ,

$$\mathbf{y}_i \mid \boldsymbol{\theta}_i, \mathbf{x}_i \sim \text{Adj.Multinomial}(N_i, \boldsymbol{\theta}_i = (\theta_{i,1}, \dots, \theta_{i,K}))$$

$$\boldsymbol{\theta}_i \mid \boldsymbol{\alpha} \sim \text{Dirichlet}(\alpha_1, \dots, \alpha_K) \quad (\text{prior})$$

$$\boldsymbol{\theta}_i \mid \mathbf{y}_i, \mathbf{x}_i \sim \text{Dirichlet}(\boldsymbol{\alpha} + \mathbf{y}_i) \quad (\text{posterior})$$

- ▶ Each terminal node's count vector $\mathbf{y}_i = (n_{i1}, \dots, n_{iK})_{K \times 1}$ across K weight bins for $i = 1, \dots, 128$ rows.
- ▶ Dirichlet prior adds *pseudo-counts* α_k to smooth zero cells (Poole and Mackworth, 2025).
- ▶ Use conjugacy $\Rightarrow$ closed-form node evidence used as impurity.

Why drop the multinomial coefficient?

- ▶ For a split $N = N_1 + N_2$, the coefficient
$$\binom{N}{n_1, \dots, n_K} > \binom{N_1}{\cdot} + \binom{N_2}{\cdot}$$
 no matter the counts.
- ▶ If kept, CART would favor *larger* partitions, not purer ones.
- ▶ Omitting it lets the criterion model solely the distributional fit.

Derivation of DM Likelihood — Key Steps

1. Joint density

$$p(\mathbf{y}, \boldsymbol{\theta} \mid \mathbf{x}) = p(\mathbf{y} \mid \boldsymbol{\theta}, \mathbf{x}) p(\boldsymbol{\theta}) = \underbrace{\prod_{k=1}^K \theta_k^{n_k}}_{\text{Adj. Multinomial}} \underbrace{\frac{1}{B(\boldsymbol{\alpha})} \cdot \prod_{k=1}^K \theta_k^{\alpha_k - 1}}_{\text{Dirichlet}} = \frac{\Gamma(\alpha_0)}{\prod_k \Gamma(\alpha_k)} \prod_{k=1}^K \theta_k^{n_k + \alpha_k - 1}$$
$$\propto \prod_{k=1}^K \theta_k^{(n_k + \alpha_k) - 1} \sim \text{Dirichlet}(\boldsymbol{\alpha} + \mathbf{y})$$

2. Integrate out $\boldsymbol{\theta}$

$$p(\mathbf{y} \mid \boldsymbol{\alpha}, \mathbf{x}) = \int_{\boldsymbol{\theta}} p(\mathbf{y}, \boldsymbol{\theta}) d\boldsymbol{\theta} = \frac{\Gamma(\alpha_0)}{\prod_k \Gamma(\alpha_k)} B(\mathbf{y} + \boldsymbol{\alpha}) = \frac{\Gamma(\alpha_0)}{\Gamma(N + \alpha_0)} \left(\prod_{k=1}^K \frac{\Gamma(n_k + \alpha_k)}{\Gamma(\alpha_k)} \right)$$

3. Log-likelihood (used as node deviance)

$$\ell = \log p(\mathbf{y} \mid \boldsymbol{\alpha}, \mathbf{x}) = \log \Gamma(\alpha_0) - \log \Gamma(N + \alpha_0) + \sum_{k=1}^K [\log \Gamma(n_k + \alpha_k) - \log \Gamma(\alpha_k)].$$

4. Split gain $\Delta \ell = \ell_{\text{left}} + \ell_{\text{right}} - \ell_{\text{parent}}$.

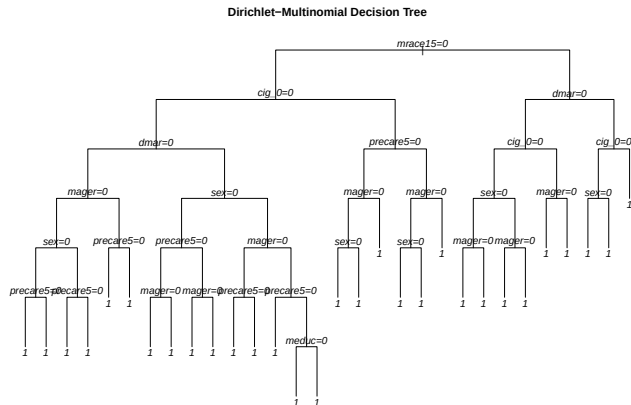
(based on Mimno, 2025; Gundersen, 2020; Minka, 2000)

DM-CART Construction

- 1 Compute $\Delta\ell$ for every candidate split.
- 2 Pick the largest positive gain; recurse until no gain or max depth.
- 3 Fit two trees: (i) full (LBW + NBW) and (ii) LBW-only.

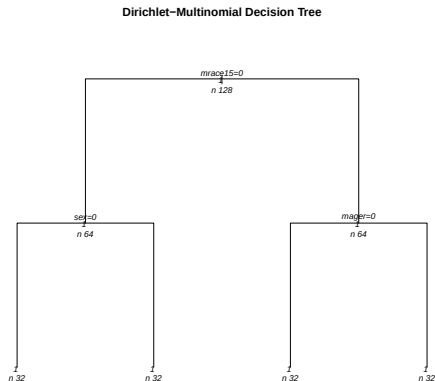
Using `rpart` (Therneau and Atkinson, 2023; Therneau et al., 2025).

Full Model Tree Structure



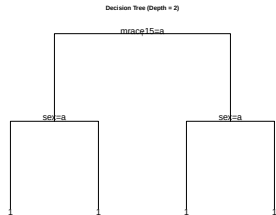
- ▶ **Root split:** maternal race (Black vs non-Black) – biggest deviance drop.
- ▶ Next: smoking status, then marital status.
- ▶ Model focus: socioeconomic/demographic

LBW-only Model Tree Structure

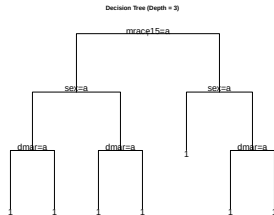


- ▶ Race still dominates the root.
- ▶ Race, infant sex, and maternal age are the only considered predictors.
- ▶ Model focus: biological/genetic

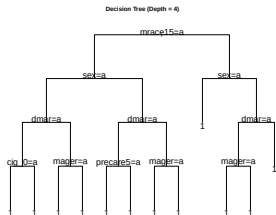
Depth Comparison — Full Model



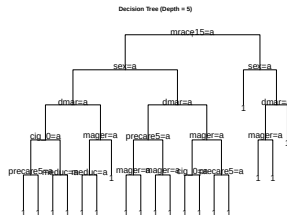
depth = 2



depth = 3



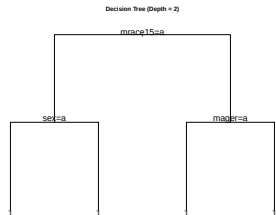
depth = 4



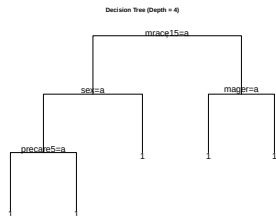
depth = 5

- ▶ Terminal nodes grow from 4 → 19.
- ▶ Splitting order: race → sex → marital remains entirely unchanged.
- ▶ Interestingly, smoking appears only at depth 4.
- ▶ For non-Black mothers, the branch is far more shallow than for Black mothers.

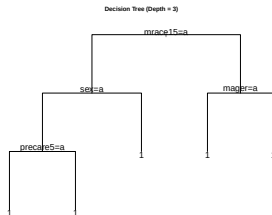
Depth Comparison — LBW-Only Model



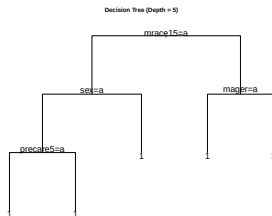
depth = 2



depth = 4



depth = 3



depth = 5

- ▶ Terminal nodes grow from 4 → 5, much shallower.
- ▶ Race still opens every tree.
- ▶ Splitting order: infant gender, maternal age, then prenatal care. Improvement stagnates quickly with restricted LBW data.
- ▶ Smoking *never* included.

Two-Tier Parametric Bootstrap Methodology ($B = 10,000$)

Notation: $T = \sum_i \sum_k n_{ik}$ (grand total), $N_i = \sum_k n_{i,k}$ (class total) and $p_i = N_i/T$ (empirical class shares).

Tier 1 — Between-class counts resampling

Draw a new vector of class totals, once per class, *propagating prevalence of N profiles*.

$$\mathbf{n}^* = (n_1^*, \dots, n_N^*) \sim \text{Multinomial}(T, \mathbf{p}), \quad i = 1, \dots, N = 128$$

Tier 2 — Within-class counts resampling

Given new $n_i^* > 0$, resample K category counts:

$$\tilde{\mathbf{y}}_i = (\tilde{n}_{i,1}, \dots, \tilde{n}_{i,K}) \sim \text{Multinomial}(n_i^*, \hat{\boldsymbol{\pi}}_i), \quad \text{where } \hat{\boldsymbol{\pi}}_i = (\hat{\pi}_{i,1}, \dots, \hat{\pi}_{i,K})$$

$\hat{\pi}_{i,k} = (n_{ik} + \alpha_k)/(N_i + \alpha_0)$ held *fixed* during resampling (focusing on observed counts variability).

$\hat{\boldsymbol{\pi}}_i$ are the *mean bootstrap probability estimates* for K birth-weight categories across B replicates. Interpreted as the best guess of a future birth of class i to fall into category k .

$\mathbb{E}[\tilde{\mathbf{y}}_i] = n_i^* \hat{\boldsymbol{\pi}}_i$ are the *expected counts* for category k in class i , centered around $\boldsymbol{\pi}_i$.

Interpreting the Ensemble

- ▶ Each bootstrap tree is fitted to the resampled $\tilde{\mathbf{Y}}$ with DM-CART.
- ▶ For every predictor we record:
 - **Variable-Inclusion percent:** proportion of trees where the variable appears (any depth).
 - **Root-split percent:** proportion of trees where variable is used as root split.
- ▶ Proportions show predictors importance under each scope using the DM-CART method.
- ▶ We also aggregate $\hat{\pi}_{i,k}$ across replicates to obtain point estimates and 95% percentile intervals for any subgroup.
- ▶ Contrast: *high-risk* (class 69) and *low-risk* (class 28) probability estimates.
 - *High-risk:* unmarried, Black, smoking mother under 33 with < High-School education, inadequate prenatal care, delivering female infants
 - *Low-risk:* married, non-Black, non-smoking mothers aged 33+, $\geq$ High-School education, adequate prenatal care, delivering male infants

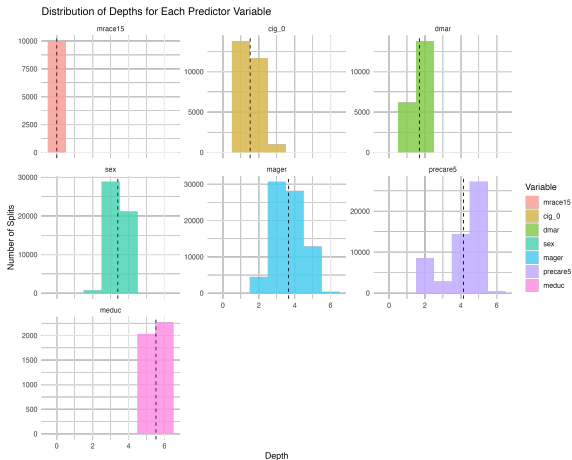
Variable Importance: Full vs. LBW-Only Models

	Full Model	LBW-Only Model
<i>Initial Split Variable</i>		
mrace15	1.0000	1.0000
<i>Variable Frequency</i>		
sex	1.0000	1.0000
dmar	1.0000	0.0915
mrace15	1.0000	1.0000
mager	1.0000	0.9741
precare5	1.0000	0.6351
cig_0	1.0000	0.0196
meduc	0.3794	0.0000

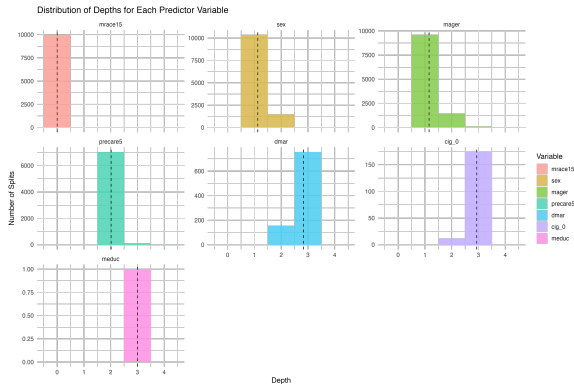
Note: "Variable frequency" (proportion of trees containing each variable) and "Initial Split Variable" (normalized measure of predictive contribution) for both models.

Predictor Depth Across Ensemble

Histograms of predictor inclusion depth across ensemble.



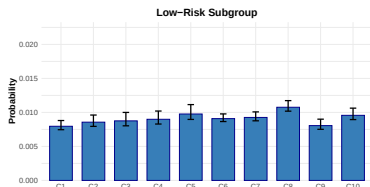
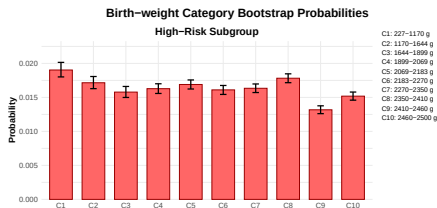
Full model



LBW-only model

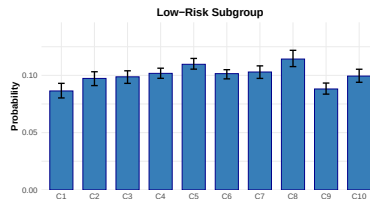
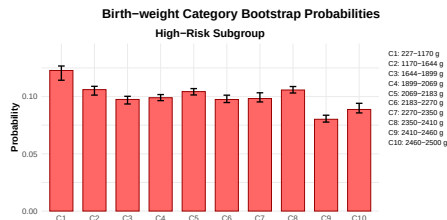
Bootstrap Probability Estimates by Risk Profile

Aggregate mean $\hat{\pi}_{i,k}$ across $B = 10,000$ bootstraps for *high-risk* (class 69) and *low-risk* (class 28) maternal profiles. Error bars show 95% percentile intervals.



Full model (without C11/NBW category):

$\pm \approx 0.002$



LBW-only model: $\pm \approx 0.008$

Predictor Importance & Risk Across 10 000 Bootstrap Trees

► Full model

- Race, smoking, marital status selected in every tree and dominate root splits.
- Prenatal care & maternal age enter later; education appears in 38% of trees at deep nodes.
- High-risk vs. low-risk NBW estimates reduced by 7% (83.6% vs. 90.9%).
- Clear disadvantage in high-risk profile. For most severe BW category, probability estimates are more than twice as large (1.9% vs. 0.8%)

► LBW-only model

- Race and infant gender selected in every tree; maternal age in 97% of trees.
- Socioeconomic predictors (marital, smoking, education) rarely/never chosen (<10%).
- High-risk profile reflect higher severe LBW estimates, while low-risk profile reflect less severe estimates.

- ▶ **DM-CART integration:** Dirichlet–Multinomial impurity brings Bayesian smoothing into CART, eliminating zero-cell counts while retaining strong interpretability.
- ▶ **Informed priors from 2020 data:** quantile information for binning the 2021 data creates stable and informed Dirichlet prior based on historic data.
- ▶ **Two-tier parametric bootstrap:** ensures variability in the consolidated counts data.

Conclusions

- ▶ **Model focus:** Socioeconomic/demographic information used in full model to contrast diverse BW outcomes. LBW-only model shifts focus on biological/genetic predictors.
- ▶ **Key risk factors:** maternal race, smoking, and marital status dominate full model; infant sex, maternal age, and prenatal care refine LBW risk.
- ▶ **Model performance:** DM-CART delivers clear, actionable subgroup rules that align with epidemiological literature and remain robust across 10,000 bootstrap replicates.
- ▶ **Generalizability:** the Bayesian tree framework is extensible to other high-dimensional multinomial-count problems.

Limitations

- ▶ Reductionist encoding of predictor variables; discarding key information about predictors.
- ▶ Race predictor is a very coarse proxy for other socioeconomic/demographic variables. These might be a better choice depending on modeling framework.
- ▶ Limited set of seven binary predictors simplifies the problem for the sake of interpretability.
- ▶ Analysis is cross-sectional (2021) with no temporal trend.

Future Directions

- ▶ **Richer maternal-health features:** hypertension, diabetes, BMI, mental-health, prior pregnancy complications.
- ▶ **Contextual variables:** air quality, housing & food access, neighborhood crime, proximity to prenatal services.
- ▶ **Temporal modeling:** introduce multi-year trends to analyze BW outcomes.
- ▶ **Clinical deployment:** embed a risk calculator into EHR workflows; practitioner usage for clinical risk assessment.

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Thank you!

Questions?

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